MILITARY SPECIFICATION

PLOTTING INSTRUMENT SET, STEREOPLOTTER, MULTIPLEX LABORATORY, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for preservation, packaging, packing and marking of a Plotting Instrument Set, Stereoplotter, Multiplex Laboratory (SC 6675-93-CL-E10).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

R-N-91	- Naphthalene; Technical.
GG-T-35	- Tables; Drawing, Tracing, and Scribing.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.

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MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-R-11254	- Rectifier, Projection Printing, Photogrammetric: 9-1/2 Inch Aerial Roll Film; Vertical Projection, 0 to 20 Degree Tilt, with 0.7 to 3.3 Magnification at 20 Degree Tilt, Autofocusing.
MIL-S-12781	- Sink, Photographic Processing, and Sink, Lithographic Processing.
MIL-S-13565	- Sinks, Photographic Processing, Temperature-Controlled.
MIL-P-14145	- Print Washer, Photographic: Rotary Pan Type, (EK-4).
MIL-P-14257	- Printer, Reduction, Stereoplotter, Multiplex.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-G-28554	- Generator Sets, Mobile Electric Power; Packaging of.
MIL-P-52281	- Processing Machine, Photogrammetric Diapositive Plates.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein and in applicable specifications and standards. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.2 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2), to prove prior to starting production packaging and packing, that the applied preservation, packaging, packing, and marking comply with the requirements of this specification. Examination and test shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Preservation and packaging. Preservation and packaging shall be Level A, B, or C, as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Methods. Methods of preservation, indicated by symbols herein, shall conform to the corresponding symbols in MIL-P-116.

3.3.1.2 Craftsman knife and shears. Each item shall be preserved and packaged in accordance with PPP-P-40, Level A.

3.3.1.3 Aprons, paperweights, chamois leathers, photographic filters, paper towels, cheesecloth, dressmakers needles, pen points, pencil lead,

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penholders, tack claw, print rollers, wood rulers, squeegee, stapler, staples, thumbtacks, thermometers, erasers, pencil pointer, drafting triangles, pin vise, and pencils. Each item, box, or package shall be preserved Method IC-1.

3.3.1.4 Sponges and print paddles. The items of like description shall be preserved together, Method IC-1.

3.3.1.5 Processing tray. Each tray shall be preserved Method IC-2.

3.3.1.6 Trip balance, fire extinguisher and bracket, folding chair, rotary chair, tape dispenser, desk light, extension light, rectifier light, exposure meter, paper trimmer, beam indicating scales, darkroom safelight, and towel dispenser. Each item shall be preserved Method III. The containers shall conform to PPP-B-636, W5c, W6c, or V3c, as applicable to the weight of contents. The boxes shall be closed in accordance with Method V of the appendix to the box specification.

3.3.1.7 Bottle stoppers, applicator bottles, glass funnels, photographic reducer, liquid graduates, cans of developer, hydrometers, orbit bath containers, bottles of ink and thinner, fluorescent lamps, film clips, and packages of sodium carbonate. Items of like description shall be preserved together Method III. The containers and closure shall be as specified in 3.3.1.6. The weight and size limitations of the boxes shall not be exceeded.

3.3.1.8 Generator set. The generator set when furnished (MAP only) shall be preserved, and packed in accordance with MIL-G-28554, Level A. The unit container shall be a box conforming to PPP-B-601, Domestic Type, Grade B.

3.3.1.9 Exposure control instrument and interval timer. Each item shall be preserved Method IIb. The inner containers shall conform to PPP-B-636, Class Domestic, Variety SW, Grade 125, style optional and the outer containers shall conform to PPP-B-636, Grade W6c. The outer containers shall be closed as specified in 3.3.1.6.

3.3.1.10 Paint brushes. The bristles of each brush shall be encased in a wallet type envelope of minimum 60-pound basis weight Kraft or equivalent paper. The envelope shall be taped to the brush handle. The brushes shall then be preserved together Method IC-1 with three balls or an equivalent quantity of flaked naphthalene conforming to R-N-91 placed inside the barrier material.

3.3.1.11 Artists brush set. The artist brush set shall be placed in a commercial-type artist brush envelope. The envelope containing the brushes shall contain a minimum of three balls or equivalent quantity of flake naphthalene conforming to R-N-91. The envelope shall then be preserved Method IC-1.

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3.3.1.12 Photographic print driers. Electrical components of the driers shall be preserved in accordance with MIL-E-16298, Level A using the alternate method specified for equipment assembled to machines. Each drier shall be placed in a box conforming to PPP-B-601, Domestic Type, Grade B. Blocking and bracing shall be provided to prevent movement of the driers within the boxes.

3.3.1.13 Photographic print washer. The photographic print washer shall be preserved and packaged in accordance with MIL-P-14145, Level A. The unit container for the printer shall be a box conforming to PPP-B-601, Domestic Type, Grade B.

3.3.1.14 Stereoplotter printer. The stereoplotter printer shall be preserved and packaged, in accordance with MIL-P-14257, Level A. The unit container for the printer shall be a box conforming to PPP-B-601, Domestic Type, Grade B.

3.3.1.15 Photogrammetric processing machine. The photogrammetric processing machine shall be preserved, and packaged, in accordance with MIL-P-52281, Level A.

3.3.1.16 Photogrammetric rectifier. The photogrammetric rectifier shall be preserved and packaged in accordance with MIL-R-11254, Level A.

3.3.1.17 Photographic printer.

3.3.1.17.1 Technical publications. The technical publications shall be preserved Method IC-1.

3.3.1.17.2 Accessories and repair parts. Accessories and repair parts shall be preserved and packaged in accordance with the applicable provisions of MIL-P-116.

3.3.1.17.3 Printer. All surfaces of the printer requiring preservation in accordance with MIL-P-116 shall be coated with Type P-7 preservative. The preserved surfaces shall then be wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, Class 2 and the material secured with tape conforming to PPP-T-60, Type IV. The printer shall then be packaged Method IIb. The inner container shall be a box conforming to PPP-B-636, Class Domestic, Type CF, Variety DW, Grade 600 and the outer container shall conform to PPP-B-601, Domestic Type, Grade B.

3.3.1.17.4 Consolidated packaging. The technical publications and accessories and repair parts shall be packaged in a box conforming to PPP-B-636, Class Weather Resistant. Closure of the box shall be as specified in 3.3.1.6.

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3.3.1.18 Sinks. The sinks shall be preserved and packaged in accordance with the Level A requirements of MIL-S-12781 or MIL-S-13565 as applicable for the type of sink involved. The outer containers for each sink and accessories shall conform to PPP-B-601, Domestic Type, Grade B.

3.3.1.19 Photographic print straightener. The photographic print straightener shall be preserved Method IIb. The inner container shall conform to PPP-B-636, Class Domestic, Variety SW, Grade 175, style optional and the outer container shall conform to PPP-B-636, V3s. After closure, the outer container shall be closed as specified for fiber boxes in 3.3.1.6.

3.3.1.20 Tracing table. The tracing table shall be preserved and packaged, in accordance with GG-T-35, Level A. The table and components shall be placed in a box conforming to PPP-B-601, Domestic Type, Grade B.

3.3.1.21 Consolidated packaging. The 32 gallon garbage can shall be utilized for consolidated packaging of preserved and packaged small components. The sink compartments, if practical, shall also be utilized to the maximum extent possible for consolidated packaging of components. Nonabsorbent cushioning material conforming to PPP-C-843, Type II, Class A, shall be used to cushion and surround the items in a manner to prevent damage to the items or the can and sink compartments by movement of the items. The can lid shall be installed and sink compartment doors closed and secured to prevent opening during handling or shipment. Other components of the set, except 5 gallon cans, large items and items packaged in wood boxes shall be consolidated packaged in close-fitting boxes conforming to PPP-B-601, Domestic Type, Grade B. Cushioning shall be applied as specified herein. The garbage can containing packaged items shall be packaged in a box specified herein. The boxes shall be closed in accordance with the appendix to the box specification.

3.3.2 Level B. Components of the set shall be preserved and packaged as specified for Level A with the following exceptions:

(a) The generator set shall be preserved and packaged in accordance with MIL-G-28554, Level B. The unit container shall be as specified in 3.3.1.8.
(b) The photogrammetric processing machine shall be preserved and packed in accordance with MIL-P-52281, Level B.
(c) Photographic processing sinks in accordance with MIL-S-13565, shall be preserved and packaged, Level B of MIL-S-13565.
(d) Items to be preserved Methods IIa or IIb of Level A shall be preserved Methods IA-16 or IA-14, as applicable.

Consolidated packaging shall be as specified for Level A.

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3.3.3 Level C. The components of the set shall be preserved and packaged to afford protection against corrosion, deterioration, and physical damage from the supplier to the initial destination. The supplier may use his own materials and commercial practice provided they fulfill these requirements.

3.4 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

3.4.1 Level A.

3.4.1.1 Photogrammetric processing machine. The photogrammetric processing machine, preserved and packaged as specified in 3.3 shall be packed in accordance with MIL-P-52281, Level A.

3.4.1.2 Photogrammetric rectifier. The rectifier, preserved and packaged as specified in 3.3 shall be packed in accordance with MIL-R-11254, Level A.

3.4.1.3 Other components. The remaining components of each set, preserved and packaged as specified in 3.3, shall be packed together in two or more crates, of approximate weight and cube, not to exceed four crates. Crates shall conform to MIL-C-104, Type I, Class 1 or 2, Style a. Blocking, bracing, and anchoring of contents shall be in accordance with the crate specification and MIL-STD-1186. After closure, the crates shall be strapped in accordance with the crate specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable; and, unless otherwise specified, shall be Finish B (see 6.2). When specified, Finish A strapping shall be used.

3.4.2 Level B.

3.4.2.1 Photogrammetric processing machine. The photogrammetric processing machine, preserved and packaged as specified in 3.3 shall be packed as specified in MIL-P-52281, Level B.

3.4.2.2 Other components. The remaining components of each set, preserved and packaged as specified in 3.3, shall be packed together in two or more crates, not to exceed four crates. Crates shall conform to MIL-C-104, Type I, Class 1 or 2, Style a, as specified for Level A; or crates conforming to MIL-C-3774, nailed assembly. Blocking, bracing, and anchoring of contents, and strapping of the crates shall be as specified for Level A, except strapping need not be zinc coated.

3.4.3 Level C. The components of the set, preserved and packaged as specified in 3.3, shall be prepared for shipment in a manner to assure carrier

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acceptance and safe delivery to destination at the lowest ratings of the carrier. Containers as a minimum shall comply with Uniform Freight Classification rules or with National Motor Freight Classification rules.

3.5 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), packages and shipping containers shall be marked in accordance with MIL-STD-129.

3.6 Workmanship. Workmanship shall be such as to provide maximum protection to the set components, by application of the requirements specified herein, to prevent corrosion, deterioration, and mechanical or physical damage to the components during handling, shipment, and storage, and shall be in accordance with the highest grade applicable for this type of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects listed in 4.4.4. Presence of one or more defects shall be cause for rejection of the preproduction pack.

4.3.2 Test. Each crate, as packed for Level A or B, shall be subjected to the guided-impact test (railroad-car) in Appendix A to MIL-STD-1186. The

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test car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 miles and not more than 11 miles per hour. Shifting of contents, loosening or breaking of anchoring, blocking, or bracing within the crate shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Inspection stages. Inspection shall be in three stages as follows:

(a) The first stage shall include inspection of procedures, methods, materials, containers, and packaging marking prior to placing the packages into consolidated containers.
(b) The second stage shall include inspection of closure and marking of consolicated containers, and the arrangement and blocking and bracing of consolidated containers in the shipping containers.
(c) The third stage shall include inspection of the closed, strapped, and marked shipping containers as prepared for shipment.

4.4.2 Unit of product. For the purpose of inspection all packages or packs, prepared for any applicable inspection stage, that will comprise a complete pack shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in Table I. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

Table I. Classification of Defects

No.	Defect	Level A	Level B	Level C
101.	Materials, methods and containers not as specified. Each incorrect material, method or container shall constitute one defect (3.1, 3.3, 3.4).	X	X	X
102.	Knife and shears not preserved and packaged in accordance with the referenced document as specified (3.3.1.2, 3.3.2).	X	X	-
103.	Generator set not preserved and packaged in accordance with the referenced documents as specified (3.3.1.8, 3.3.2).	X	X	-
104.	Brush set not placed in an envelope and envelope does not contain naphthalene as specified (3.3.1.10, 3.3.2).	X	X	-
105.	Driers not preserved in accordance with the referenced document as specified (3.3.1.11, 3.3.2).	X	X	-
106.	Washer not preserved and packaged in accordance with the referenced document as specified (3.3.1.12, 3.3.2).	X	X	-
107.	Photographic printer not preserved, packaged and packed in accordance with the referenced document as specified (3.3.1.13, 3.3.2).	X	X	-

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Table I. Classification of Defects (Con'd)

No.	Defect	Level A	Level B	Level C
108.	Stereoplotter printer not preserved and packaged in accordance with the referenced document (3.3.1.14, 3.3.2).	X	X	-
109.	Photogrammetric processing machine not preserved, packaged and packed in accordance with the referenced document (3.3.1.15, 3.3.2).	X	X	-
110.	Photogrammetric rectifier not preserved, packaged, and packed in accordance with the referenced document (3.3.1.16, 3.3.2).	X	X	-
111.	Sinks not preserved and packaged in accordance with the referenced document as specified (3.3.1.17, 3.3.2).	X	X	-
112.	Tracing table not preserved and packaged in accordance with the referenced document (3.3.1.19, 3.3.2).	X	X	-
113.	Consolidated packaging not as specified (3.3.1.20, 3.3.2).	X	X	-
114.	Fiberboard boxes not closed as specified (3.3.1.6, 3.3.1.7, 3.3.1.9, 3.3.1.17.4).	X	X	-
115.	Strapping on crates not zinc coated as specified (3.4.1).	X	-	-
116.	Marking illegible, incomplete, or incorrect (3.5).	X	X	X

5. PREPARATION FOR DELIVERY

(This section not applicable to this specification.)

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts or purchase orders for the preservation, packaging, packing, and marking of a Plotting Instrument Set, Stereoplotter, Multiplex Laboratory (NSN 6675-00-641-3621) or MAP (NSN 6675-00-926-1245).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction pack (see 3.2).
(c) Level of preservation and packaging and level of packing required (see 3.3 and 3.4). Level B preservation and packaging is intended to provide economical but limited protection and should be specified only when it has been determined the set components will be

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held in covered storage for an indefinite period from date of initial preservation and packaging.

(d) When other than Finish B strapping is required (see 3.4.1.3).

(e) Any special marking required (see 3.5).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the set components in accordance with this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 6675-A151

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE MIL-P-52650A(ME) Plotting Instrument Set, Stereoplotter, Multiplex Laboratory, Packaging Of

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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